

Probability

1. [June 2023]

A certain book has 250 pages, and 21 of these pages have an illustration. If one of the book's pages is selected at random, what is the probability of selecting a page with an illustration? (Express your answer as a decimal or fraction, not as a percent.)

2. [March 2024]

At a convention center, there are a total of 325 visitors. Each visitor is located in either room A , room B , or room C . If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.48, and the probability of selecting a visitor who is located in room B is 0.16. How many visitors are located in room C ?

- A) 25 C) 104
B) 36 D) 117

3. [March 2024]

There are 20 marbles in the bag: 8 red marbles, 10 green marbles, and 2 blue marbles. If one of these marbles is selected at random, what is the probability of selecting a red marble?

- A) $\frac{2}{20}$
B) $\frac{8}{20}$
C) $\frac{10}{20}$
D) $\frac{12}{20}$

4. [May 2024]

A choir of 48 singers has 12 members who sing bass. If one singer from the choir is selected at random, what is the probability of selecting a singer who sings bass?

- A) $\frac{1}{48}$ C) $\frac{36}{48}$
B) $\frac{12}{48}$ D) $\frac{48}{48}$

5. [May 2024]

The table shows the distribution of people in a certain city by age group.

Age group	Proportion
Less than 18 years old	27%
18–40 years old	24%
41–65 years old	28%
Greater than 65 years old	21%

If a person in this city is selected at random, which of the following is closest to the probability of selecting a person who is greater than 65 years old, given that the person is at least 18 years old?

- A) 0.21 C) 0.35
B) 0.29 D) 0.41

6. [May US 2024]

At a convention center, there are a total of 375 visitors. Each visitor is located in either room A , room B , or room C . If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.48, and the probability of selecting a visitor who is located in room B is 0.32. How many visitors are located in room C ?

- A) 20 C) 60
B) 58 D) 75

7. [December US 2024]

A sleep study consisted of 59 participants, of which 53 participants each had an average of more than 120 minutes of rapid eye movement (REM) sleep per night. If a participant from this study is selected at random, what is the probability of selecting a participant that had an average of more than 120 minutes of REM sleep per night?

- A) $\frac{53}{100}$ C) $\frac{53}{59}$
B) $\frac{59}{100}$ D) $\frac{59}{53}$

8. [May US 2024]

At a convention center, there are a total of 375 visitors. Each visitor is located in either room A , room B , or room C . If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.48, and the probability of selecting a visitor who is located in room B is 0.32. How many visitors are located in room C ?

- A) 20 B) 58 C) 60 D) 75

9. [August US 2023]

The table summarizes the distribution of age and assigned group for participants in a study.

	0-9 years	10-19 years	20+ years	Total
Group A	14	19	27	60
Group B	27	7	26	60
Group C	19	34	7	60
Total	60	60	60	180

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age? (Express your answer as a decimal or fraction, not as a percent.)

10. [August 2024]

A bin contains a mixture of T-shirts for two sports teams. The table shows the number of T-shirts in the bin, classified by size and sports team.

	Flames	Panthers	Total
Small	6	9	15
Medium	24	33	57
Large	18	24	42
Total	48	66	114

One T-shirt from the bin will be selected at random. What is the probability of selecting a T-shirt that is medium, given that the T-shirt is a Flames T-shirt? (Express your answer as a decimal or fraction, not as a percent.)

11. [June 2024]

The table summarizes the distribution of age and assigned group for 105 participants in a study.

	0-9 years	10-19 years	20+ years	Total
Group A	5	19	11	35
Group B	4	8	23	35
Group C	26	8	1	35
Total	35	35	35	105

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age

A) $\frac{2}{7}$

B) $\frac{3}{7}$

C) $\frac{19}{35}$

D) $\frac{6}{7}$

12. [October 2024]

	Live north of Center St.	Live south of Center St.	Total
Less than 45 years old	13	12	25
At least 45 years old	22	88	110
Total	35	100	135

The table summarizes members of a local organization by age and whether they live north or south of Center St. If a member of the organization is selected at random, what is the probability that the selected member is at least 45 years old?

A) $\frac{25}{135}$

B) $\frac{35}{135}$

C) $\frac{100}{135}$

D) $\frac{110}{135}$

13. [August US 2023]

A researcher calculated the mitochondrial genome sizes g , in kilobase pairs (kbp), and the number of mitochondrial genes for 34 different species. The table summarizes these data.

Number of mitochondrial genes	Mitochondrial Genome Size (kbp)				Total
	$g < 16$	$16 \leq g < 18$	$18 \leq g < 20$	$g \geq 20$	
36 or less	3	0	0	0	3
37	1	19	0	0	20
38 or more	2	0	4	0	6
Total	6	19	4	0	29

One of these species is selected at random. What is the probability of selecting a species with a mitochondrial genome size greater than or equal to 18 kbp, given that it has 37 or more mitochondrial genes?

(Express your answer as a decimal or fraction, not as a percent.)

14. [November US 2024]

For 100 neurons, the table summarizes the distribution of classification and cell body diameter:

Classification	Cell body diameter (micrometers)		
	Less than 20	20 to 30	Greater than 30
Sensory neuron	13	7	3
Motor neuron	0	15	18
Interneuron	10	34	0

One of these neurons will be selected at random. What is the probability of selecting a neuron with a cell body diameter that is less than or equal to 30 micrometers, given that it is not classified as a motor neuron?

15. [November 2024]

	Even integers	Odd integers	Total
List A	12	44	56
List B	40	4	44
Total	52	48	100

The given table shows the number of even and odd integers contained in list A and list B. No integer is contained in both lists. What fraction of even integers represented in the table are in list A?

A) $\frac{12}{52}$

B) $\frac{12}{56}$

C) $\frac{52}{100}$

D) $\frac{52}{56}$